constantly changing conditions, but we can sample from the environment, and these samples at particular times are the state information of the environment we give to the agent.

- An action is a decision made by an agent that produces a change in its environment. Moving a particular chess piece is an action, and so is pressing the gas pedal in a car.
- A reward is a positive or negative signal given to an agent by the environment after it takes an action. The rewards are the only learning signals the agent is given. The objective of an RL algorithm (i.e., the agent) is to maximize rewards.
- The general pipeline for an RL algorithm is a loop in which the agent receives input data (the state of the environment), the agent evaluates that data and takes an action from a set of possible actions given its current state, the action changes the environment, and the environment then sends a reward signal and new state information to the agent. Then the cycle repeats. When the agent is implemented as a deep neural network, each iteration evaluates a loss function based on the reward signal and backpropagates to improve the performance of the agent.